Homework 7: Scheduling Report
CS231P
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1.

Time/PE	π1	π2	π3	π4	π5
1	1	2	1	1	2
2	1	3	5	3	2
3	3	4	5	5	3
4	5	5			5
5	6				5
6					6

Explain why your proposed temporal schedule is legal?
No VP of a job is ahead of any other VP of the same job by more than one global communication step at any time. There is not more than 1 step of communication between the VPs for any of the 6 jobs.
Indicate the number of cycles in its period.
There are 6 cycles in its period.
Compute the idling ratio of the schedule.
Idling ratio = idle/total = 9/30

2.

Time/PE	π1	π2	π3	π4	π5
1	1	2	1	1	2
2	1	3	5	3	2
3	3	4	5	5	3
4	6	5	1	1	5
5	5		1		6
6			1		5

Total cycles= 6 cycles
Idle slots=5
Idling ratio = idle/total = 5/30

To lower the idling ratio, we assigned the VPs of Job 1 to processors 3 and 4 after they finished their tasks. Since we assigned Job 1 to processors 3 and 4 at time 4, we had to switch the order that processors 1 and 5 completed Job 5 and 6, so that the trailing VPs were chosen before any non trailing VP.

3. Is there a best periodic temporal schedule?

Yes, there is a best periodic temporal schedule, with an idling ratio of 0/30.

Time/PE	π1	π2	π3	π4	π5
1	1	2	1	1	2
2	1	3	5	3	2
3	3	4	5	5	3
4	6	5	1	1	5
5	5	4	1	3	6
6	3	3	1	3	5

Idling ratio: 0/30

Improving on our solution for number two to minimize the idling ratio, we filled in the idle spaces with VPs from Jobs 3 and 4. It is still legal as no VPs of jobs 3 or 4 are ahead of any other VP of the same job by more than one global communication step, and any trailing VPs are chosen before any non trailing VPs.